

Enough for Both: AI, Water and Where Data Centres Are Built

Why location matters more than volume

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Abstract

Claims that artificial intelligence is draining the world's water have spread widely online. This article summarises the evidence on that question, which is examined in full, with its sources, calculations and limitations, in the research paper *A Fixed Claim on a Shrinking Supply: AI Data Centres, Water Stress and Lock-In* (Vondoom 2026). Data centres, including those running AI, account for about 0.14% of the fresh water humanity withdraws each year; farming, including livestock, and leaking water pipes each account for far more. The risk lies elsewhere: many data centres are being built in regions already short of water, and their location fixes their water source for decades. Using Egypt as an example, the article shows that the means to avoid this already exist, and outlines three measures: non-drinking water for cooling in dry regions, planning for a facility's whole life, and using AI to reduce water lost from leaking networks.

Keywords: artificial intelligence; data centres; fresh water; water scarcity; Egypt; Nile; desalination

In short

AI does use water, but a tiny share of it. All the world's data centres together withdraw about 0.14% of the fresh water humanity takes each year. Farming takes 72%, and leaking water pipes lose about 23 times more than every data centre combined.

The real risk is not how much water AI uses, but where its data centres are built. Many are going up in places that are already running short of water, and once built, they stay for decades.

The solutions already exist and are affordable. Data centres do not need drinking water to stay cool, and the technology to use seawater or recycled water is already working. What is missing is the rule that makes it standard.

Does AI "drink" the world's water?

In 2024, an analysis by the *Washington Post* and researchers at the University of California, Riverside estimated that writing a single 100-word email with a chatbot used about half a litre of water, roughly one bottle. The image stuck, and the idea that AI is drinking the world's water has spread widely online.

Other estimates are very different. In 2025, Google measured its own AI system in use and put the water for a typical text request at 0.26 millilitres, a few drops. The two figures do not necessarily contradict each other: they measure different things. The higher one includes the water used by power stations to produce the electricity; the lower one counts only the water used at the data centre itself.

Neither number, on its own, says whether AI is causing water shortages. To answer that, AI has to be compared with everything else people use water for.

How much fresh water is there?

Almost all of Earth's water, about 97.5%, is salt water in the oceans. Of the 2.5% that is fresh, more than two-thirds is frozen in ice caps and glaciers and most of the rest is deep underground. The part people can realistically use, in rivers, lakes and shallow groundwater, is less than 1% of all fresh water.

Fresh water is not a tank that simply empties. Rain and snow refill rivers and lakes every year, and humanity currently takes roughly a third of the renewable supply that can be reached. So there is no single date when the world "runs out". Shortages are local: a quarter of the world's population lives in 25 countries that already use more than 80% of their available water every year. And some sources do behave like tanks that are being drained. Over a third of the world's monitored underground water sources are falling by more than 10 centimetres a year, and glaciers, which supply water to about 1.9 billion people, are shrinking.

Where the water actually goes

Humanity takes about 4,000 cubic kilometres of fresh water a year. Most of it goes to growing food (Figure 1).



Figure 1. Global freshwater withdrawals by sector, 2021. Data: UN World Water Development Report 2025, from FAO AQUASTAT.

Within farming, livestock is the biggest user. The best-known figure is that a kilogram of beef takes about 15,400 litres of water, but about 94% of that is rain falling on grass and crops, water that would never have reached a tap anyway. Counting only water taken from rivers, lakes and underground, livestock farming still uses about 150 cubic kilometres a year, roughly 270 times what all data centres consume.

A large amount is also simply lost. Water pipes around the world leak about 126 cubic kilometres of cleaned, treated water every year, almost 30% of the water utilities produce. In Singapore the figure is 4%; in some countries it is over 80%.

Divided by the world's population, the picture looks like this (Figure 2).

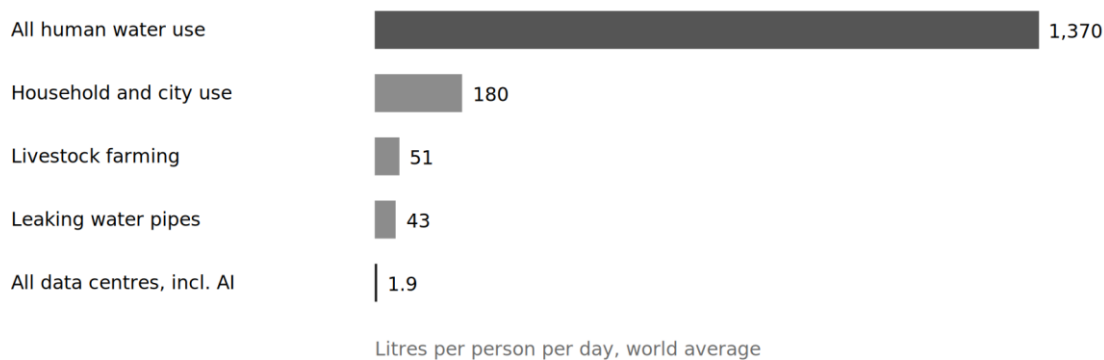


Figure 2. Daily water use per person, world average. Livestock is water taken from rivers, lakes and underground only. Data centres are 2023 figures. Source: calculated in the research paper from UN, FAO, IEA and peer-reviewed data.

On a global scale, the claim that AI is draining the world's water does not hold up. Data centres use about 2 litres per person per day out of 1,370.

So is there nothing to worry about?

There is, but it is about location, not totals. Water is used where it is found. A single large data centre can use as much water every day as thousands of households. In a region with plenty of water, that hardly matters. In a region that is already short, it competes directly with homes and farms.

That is where much of the growth is happening. About half of the world's data-centre hubs are in areas under high water stress, and in the United States about two-thirds of new data centres built since 2022 are in such places. A recent study of ten US sites found that a data centre's demand ranged from 0.2% of the local water system's capacity to more than all of it. The global share can be tiny while the local strain is severe.

The reason is simple. Data centres are built where land and electricity are cheap and taxes are low. Water rarely enters the decision, because it is rarely priced at what it is really worth. In southern Nevada, the water authority itself said that water-cooled systems had spread because water was cheaper than electricity, and in 2023 it banned that kind of cooling in new commercial and industrial buildings.

Why the choice of location lasts so long

A data centre is designed to run for 15 to 20 years. Its cooling system can be upgraded during that time, but its location cannot. A facility built far inland in a dry region can never switch to cooling with seawater, and one built far from a wastewater plant cannot easily switch to recycled water. The decision made today about where to build fixes which water sources are available until the mid-2040s. By 2050, the entire population of the Middle East and North Africa is projected to live under extremely high water stress.

Egypt: a case in point

More than 98% of Egypt's renewable water comes from outside its borders, almost all of it through the Nile. The country needs about 123 billion cubic metres of water a year but has a renewable supply of about 57 billion. Each Egyptian now has about 472 cubic metres a year, below the 500 cubic metres that marks "absolute scarcity". In 1947 the figure was 2,526 (Figure 3). Egypt's share of the Nile, set by a 1959 agreement, has not changed since.

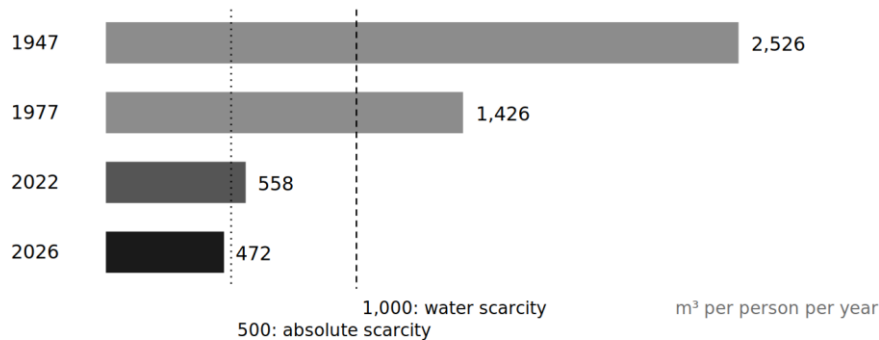


Figure 3. Renewable water per person in Egypt. Sources: CAPMAS (1947), via climate-diplomacy.org; MEED (2024), for 1977 and 2022; Ministry of Water Resources and Irrigation (2026). Sources use differing methods; the trend, not the exact values, is the point of comparison.

Upstream, Ethiopia opened the Grand Ethiopian Renaissance Dam in 2025, with a reservoir that can hold 74 billion cubic metres, and in 2026 announced three more dams on the Blue Nile. Ethiopia needs the electricity: almost half its population has none. Egypt's concern is that there is no binding agreement on how the water is released during long droughts. The research paper takes no side in this dispute. It uses the Nile as the clearest example of a large population depending on a single water source with no agreed rule for sharing it under pressure.

At the same time, Egypt is actively seeking data-centre investment. In 2026 it licensed a new data-centre operator with plans worth about US\$400 million, discussed a large data-centre and AI zone with international developers, and reviewed plans for an 80-megawatt data centre in the Suez Canal Economic Zone.

One 80-megawatt data centre cooled with evaporating water would use about 1,600 cubic metres a day; a large 1-gigawatt campus, about 20,000. That is a tiny share of Egypt's national supply. The concern is the direction: if the first facilities are allowed to run on Nile water, that becomes the model for everything that follows, for decades.

Egypt also has what it needs to avoid that. It already has capacity to turn about 1.3 million cubic metres of seawater a day into fresh water, and plans to reach 8.8 million by 2050. It is already tendering new seawater plants in the same economic zone where data centres are planned. And its Benban solar park produces more than 4 billion kilowatt-hours of electricity a year.

A pattern worth taking seriously

Reported cases of violence connected to water have risen sharply, from 24 in 2000 to 420 in 2024, most of them local disputes over access. Between countries, cooperation over shared rivers is still far more common than conflict. Research on shared rivers has found that whether there are agreements and institutions in place to manage water matters as much as how scarce it is, or more. Tension rises when things change faster than those rules can keep up.

Data centres do not cause conflict. But large, long-lasting new demands for water, added quickly to regions without agreed rules for sharing it, are exactly the kind of change those studies describe. Problems like this are easiest to solve while there is still some room in the system. Once a crisis arrives, attention and money go to managing the emergency, and long-term fixes are pushed aside.

This has happened before. Around 2400 to 1700 BCE, farming in southern Mesopotamia declined as irrigation left salt in the soil, although historians still debate how much this contributed. What is different now is the speed and scale at which today's choices are fixed in place.

What can be done

None of the solutions is experimental. Each one is already working somewhere. The research paper proposes three measures that work together.

Measure	What it means	Already happening
1. No drinking water for cooling in dry regions	Data centres in water-stressed areas cool with recycled wastewater, seawater, or seawater turned fresh using solar or wind power, and buy electricity from sources that use little water	Google's data centre in Georgia (USA) runs on recycled water; its Finnish site has been cooled with seawater since 2011
2. Plan for the whole life of the building	Before approval, check whether the water supply will still be there in 20 years, not just today, and make operators report how much water they use and where it comes from	Singapore sets water-efficiency targets for data centres; southern Nevada has banned water-evaporating cooling in new commercial and industrial buildings
3. Use AI to fix the water system	Operators help local water utilities find and repair leaks with sensors and AI, saving at least as much water as they draw from the area	AI-based leak detection is in use and development worldwide; Egypt's desalination plan already includes AI plant management

Table 1. The three measures proposed in the research paper.

The cost is small. Supplying the cooling water of every data centre in the world from seawater would use about 0.14% of the electricity those data centres are expected to use in 2030, roughly a third of what Benban alone produces in a year. The main thing to

manage is the salty leftover water, called brine: it must be released carefully, because coral reefs in the nearby Gulf of Aqaba have been harmed by water only 10% saltier than normal when it also carried treatment chemicals.

These measures can start simply. Every country already issues permits for large buildings. Attaching water conditions to those permits needs no new institutions, and it can be done before the first large data centres are built rather than fixed afterwards at far greater cost.

The bottom line

AI is not what is emptying the world's water supply. Farming, leaking pipes and falling underground water reserves all matter far more, and they were problems long before AI arrived. Blaming AI puts the responsibility in the wrong place.

The real problem is human: water that is shared, cheap and poorly measured, now receiving a new, fast-growing and long-lasting demand in many of the places least able to afford it. AI is here to stay. The task is to make sure it grows in a way that strengthens the water systems people depend on, so that there is enough for both.

Today there is still time, money and stability to get this right. Those are resources too, and like water, they will not always be available.

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